- During starting, the cooling system should reject only less amount of heat.
- It should transfer the maximum (i.e. sufficient) amount of heat from the engine combustion chamber.

7.1 AIR COOLING SYSTEM

In this type of cooling system, the engine cylinder should be kept in such a way that more air should be in contact with the outer region of the cylinder block. Cooling fins are provided on the outer wall of the cylinder so that the contact surface areas for heat transfer to be more. These arrangements are mostly seen in all motor cycles, scooters and in small engines. Figure 7.1 shows the air cooling system.

When the vehicle moves forward, the air passes over the fins and removes the heat on the engine components and hence the engine gets cooled. In some large engines, blower arrangement is made so that the blower sucks the air from the surrounding air and blows on the surface of the cooling fins of the engine.

The efficiency of the air cooling system depends on the following characteristics,

- The speed of the air which flows on the fins
- Area of the fins that is in contact with air
- Thermal conductivity of the cylinder walls

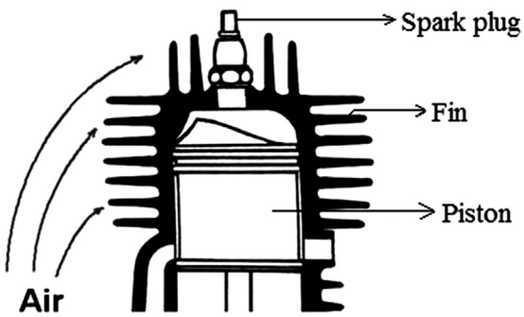


Figure 7.1 Air cooling system

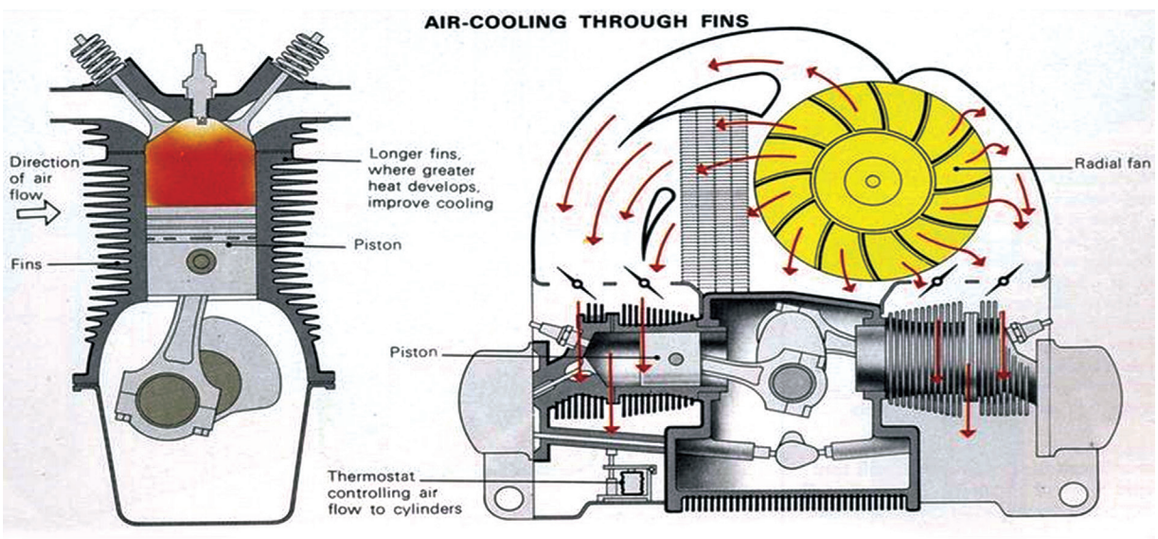


Figure 7.1(a) Air cooling system

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